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ICS 2302 SOFTWARE ENGINEERING 1

This presentation is intended to covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green Exercise tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 5

Project Management

Learning Outcomes

A study of this lesson should enable students to:

- Understand project management
- Describe Risk management
- Describe the Risk management stages

The need for management is an important distinction between professional software development and amateur programming. We need software project management because professional software engineering is always subject to budget and schedule constraints. These are set by the organization developing the software. The software project manager's job is to ensure that the software project meets these constraints and delivers software which contributes to the business goals.

5.1. Management activities

It is impossible to write a standard job description for a software manger. However, most managers take responsibilities at some stage for some or all of the following activities:

- proposal writing

- project costing
- project monitoring and reviews
- personnel selection and evaluation
- report writing and presentations

The proposal describes the objectives of the project and how it will be carried out. It usually includes cost and schedule estimates. It may justify why the project contract should be awarded to a particular organization or team.

Project planning is concerned with identifying the activities milestones and deliverables produced by a project. A plan must then be drawn up to guide the development towards the project goals.

Project cost estimation is a related activity that is concerned with estimating the resources required to accomplish the project plan. Project monitoring is a continuing project activity. The manager must keep track of the progress of the project and compare actual and planned progress and costs.

Project managers usually have to select people to work on their project. The manager is usually responsible for reporting on the project to both the client and contract organizations. Project managers must write concise, coherent documents which abstract critical information from detailed project reports.

They must be able to present this information during progress reviews consequently, the ability to communicate effectively both orally and in writing is an essential skill for a project manager.

Project Planning

Effective management of a software project depends on thoroughly planning the progress of the project. The project manager must anticipate problems which might arise and prepare tentative solutions to those problems. A plan, drawn up at the start of a project, should be used as the driver for the project. A structure for a software development plan is described below.

The Project Plan

The project plan sets out the resources available to the project, the work breakdown and schedule for carrying out the work. The details of the project plan vary depending on the type of the project and organization. However, most plans should include the following sections.

1. Introduction – this briefly describes the objectives of the project and sets out the constraints (e.g. budget, time e.t.c) which affect the project management.
2. Project organization – this describes the way in which the development team is organized, the people involved and their roles in the team.
3. Risk analysis – this describes project risks, the likelihood of the risks arising and the risk reduction strategies which are proposed.
4. Hardware and Software Resource requirements – this describes the hardware and the support software required to carry out the development. If hardware has to be bought, estimates of the prices and the delivery schedule should be included.

5. Work breakdown – this describes the breakdown of the project into activities and identifies the milestones and deliverables associated with each activity.
6. Project schedule – this describes the dependences between activities, the estimated time required to reach each milestone and the allocation of people to activities.
7. Monitoring and reporting mechanisms – this describes the management reports which should be produced and the project monitoring mechanisms used.

The project plan should be regularly revised during the project. Some parts, such as the project schedule, will change frequently; other parts will be more stable. A document organization which allows for the straight forward replacement of sections should be used.

Milestones and Deliverables

Managers used information. As software is intangible, this information can only be provided as documents that describe the state of the software being developed without this information, it is impossible to judge progress and cost estimates and schedules cannot be updated.

When planning a project, a series of milestones should be established, where a milestone is an end point of a software process activity. At each milestone, there should be a formal output such as a report that can be presented to management. A deliverable is a project result that is delivered to the customer. It is usually delivered out the end of some major project phase such as specification, design etc. To establish milestones the process is broken down into basic activities with associated outputs.

5.2. Project Scheduling

Project scheduling is a particularly demanding task for software managers. Managers estimate the time and resources required to complete activities and organize them in a coherent sequence.

Project scheduling (5.1) Involves separating the total work involved in a project into separate activities and judging the time required to complete these activities.

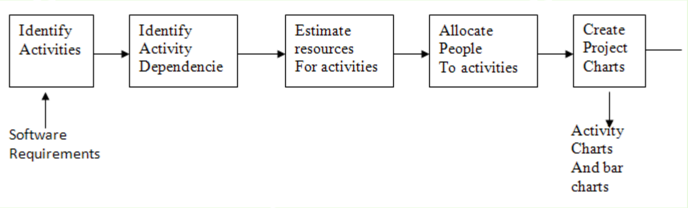

Figure 5.1:

Usually some of these activities are carried out in parallel. Project schedulers must coordinate these parallel activities and organize the work so that the workforce is used optimally. They must avoid a situation where the whole project is delayed because a critical task is unfinished.

The project schedule is usually represented as a set of charts showing the work breakdown, activities dependencies and staff allocations. Software management tools, such as Microsoft project, are now usually used to automate chart production.

Risk Management

An important task of a project manager is to anticipate risks which might affect project schedule or the quality of the software being developed and to take action to avoid these risks. The results of the risk analysis should be documented in the project plan along with an analysis of the consequences of a risk occurring. Identifying risks and drawing up plans to minimize their effects on the project is called risk management.

The categories of risks can be defined as follows:

- Project risks are risks which affect the project schedule or resources.
- Product risks are risks which affect the quality or performance of the software being developed.
- Business risks are risks which affect the organization developing or procuring the software.

The process of risk management is illustrated in figure 5.2 below:

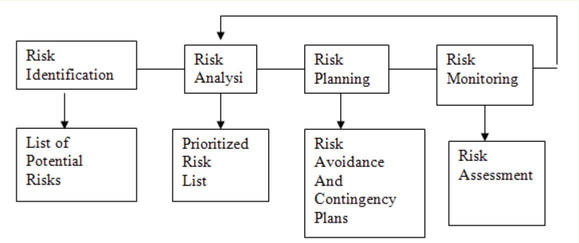

Figure 5.2:

It involves several stages:

1. Risk identification possible project, product and business risks are identified
2. risk analysis the likelihood and consequences of these risks are assessed

3. Risk planning plans to address the risk either by avoiding it or minimizing its effects on the project are drawn up
4. Risk monitoring the risk is constantly assessed and plans for risk mitigation are revised as more information about the risk becomes available.

The risk management process like all other project planning is an iterative process which continues throughout the project.

Risk Identification

This is the first step of risk management. It is concerned with discovering possible risks to the project. Risk identification may be carried out as a team process using a brainstorming approach or may simply be based on manager's experience.

Risk Analysis

During the risk analysis, each identified risk is considered in turn and a judgment made about the probability and the seriousness of the risk. There is no easy way to do this it relies on the judgment and experience of the project a manager. These should not generally be precise numeric assessments but should be based around a number of bands.

1. The probability of the risk might be assessed as very low (<10%), low (10-25%), Moderate (25-50%), high (50-75%) or very high (>75%).
2. the effect of the risk might be assessed as catastrophic, serious, tolerable or insignificant.

The results of this analysis should be tabulated in the table ordered according to seriousness of the risk. Once the risk have been analyzed and ranked, a judgment must then be made about which are the must important risks which must be considered during the project. The judgment must depend on a combination of the probability of the risk arising and the effects of that risk. In general, all catastrophic risks should be considered, as should all serious risks which have more than a moderate ability of occurrence.

Risk Planning

The risk planning process considers each of the key risks which have been identified and identifies strategies to manage the risk. The strategies fall into three categories:-

1. Avoidance strategies following these strategies means that the probability that the risk will arise will be reduced.
2. Minimization strategies following
3. these strategies means that the impact of the risk will be reduced. Contingency plans following these strategies means that, if the worst happens, you prepared for it and have a strategy in place to del with it.

Risk Monitoring

Risk monitoring involves regularly assessing each of the identified risks to decide whether or not that risk is becoming more or less probable and whether the effects of the risk have changed. Risk monitoring should be a continuous process and, at every management progress review, each of the key risks should be

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considered separately and discussed by the meeting.

To help the process a list of possible risk types may be used. These types include:

1. Technology risk which derive from software or hardware technologies which are being used as part of the system being developed.
2. People risk - associated with the people in the development team.
3. Organizational risks - risks which derive from the organizational environment where the software is being developed
4. requirements risks - risks which derive from changes to the customer of managing the requirements change
5. Tools risks - risks which derive from the CASE tools and

other support software used to derive the system.

6. Estimation risks - risks which derive from the management estimates of the system characteristics and the resources required to build the system.

Revision Questions

Example ✎. What is Project Management?

Solution: Project Management is the collection and app lication of skills, knowledge, processes, and activities to meet a specific objective that may take the form of a product or service. Project Management is an integrated process of applying 5 major processes and their related activities throughout a project lifecycle:

Initiating

Planning

Executing

Monitoring and Controlling

Closeout □

EXERCISE 1. ✍ What is a Risk?

References and Additional Reading Materials

1. Hans van Vliet (2005). Software Engineering: Principles and Practice (2nd ed.). John Wiley & Sons, Inc. ISBN: 0-471-97508-7
2. Ian Sommerville, Pete Sawyer(2005). Requirements Engineering: A Good Practice Guide. ISBN: 0-471-97444-7. John Wiley & Sons, Inc
3. Gerald Kotonya, Ian Sommerville (1998). Requirements Engineering: Processes and Techniques. John Wiley & Sons, Inc. ISBN: 0-471-97208-8

Solutions to Exercises

Exercise 1.

A Risk is an uncertain event that may hav e a positive or negative impact on the project objective. If and when it occurs, a Risk can be an opportunity (positive impact) or a threat (negative impact). There are two types of Ri sks: Business and Insurable. Risks can be grouped into various categories (Functional, Technical, Financial, and etc.)

Exercise 1